

Problem Solving and testing using Java

Week-1 Task Questions

20/07/2026-25/07/2026

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week-1 Questions:

1. Access and print the element at a given index in an array.
2. Search for a given element in a sorted array using Binary Search.
3. Find the maximum element in an array of n integers
4. Given an array of integers and a positive integer K, write a program to find :The Kth smallest element
5. Print all possible pairs of elements from an array of size n.
6. digitSum opt: sum of even or odd digits
7. Nth Fibonacci
8. Is Palindrome Number?
9. Sum of last digit of two given numbers

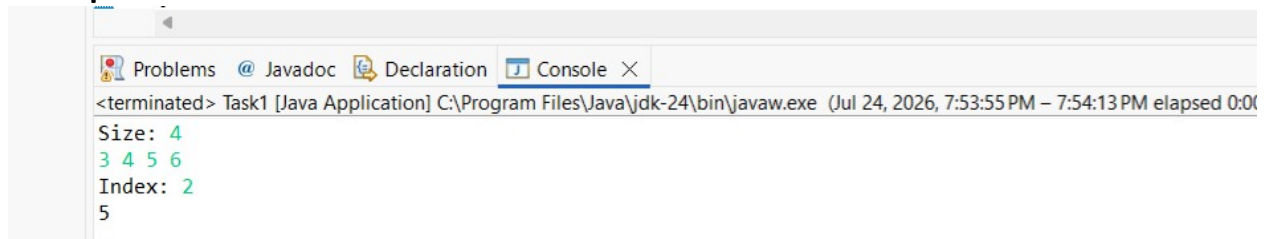
Task:1 Access and print the element at a given index in an array.

Program:

```
import java.util.*;
import java.util.stream.*;

public class Task1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Size: ");
        int n = sc.nextInt();
        List<Integer> list = IntStream.range(0, n)
            .map(i -> sc.nextInt())
            .boxed()
            .collect(Collectors.toList());
        System.out.print("Index: ");
        int index = sc.nextInt();
        Optional<Integer> value = list.stream()
            .skip(index)
            .findFirst();
        value.ifPresentOrElse( System.out::println
            ,
            () -> System.out.println("Invalid Index")
        );
    }
}
```

Output:

A screenshot of an IDE's console window. The window has tabs for 'Problems', '@ Javadoc', 'Declaration', and 'Console'. The 'Console' tab is active, showing the output of the Java application. The output text is: '<terminated> Task1 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 7:53:55 PM – 7:54:13 PM elapsed 0:0'. Below this, the program's output is displayed: 'Size: 4' followed by the numbers '3 4 5 6' on the next line, 'Index: 2' on the next line, and '5' on the final line. The numbers '3 4 5 6' and '2' are in a green color, while the others are black.

```
<terminated> Task1 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 7:53:55 PM – 7:54:13 PM elapsed 0:0
Size: 4
3 4 5 6
Index: 2
5
```

Task-2: Search for a given element in a sorted array using Binary Search.

Program:

```
import java.util.*;

import java.util.stream.*;

public class Task2 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        List<Integer> list = IntStream.range(0,n)

        .map(i->sc.nextInt())

        .boxed()

        .collect(Collectors.toList());

        Collections.sort(list);

        int key = sc.nextInt();

        int index = Collections.binarySearch(list,key);

        System.out.println(index>=0?

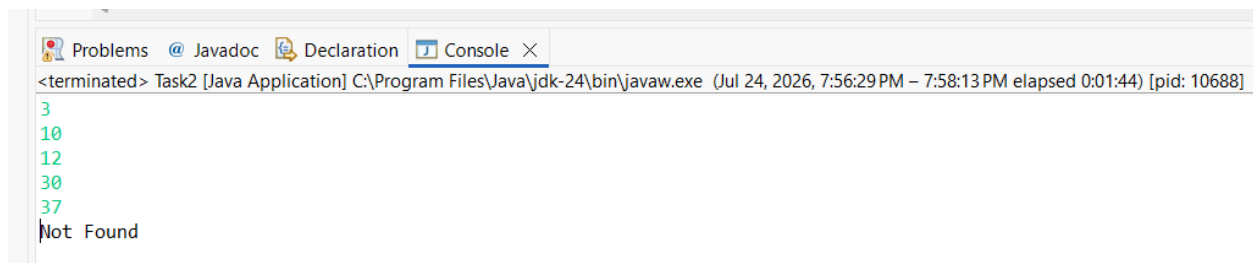
        "Found at "+index:

        "Not Found");

    }

}
```

Output:



The screenshot shows an IDE console window with the following tabs: Problems, Javadoc, Declaration, and Console. The Console tab is active, displaying the output of the program. The output consists of five lines of input numbers: 3, 10, 12, 30, and 37, followed by the text "Not Found". The console title bar indicates the application is "Task2 [Java Application]" running in "C:\Program Files\Java\jdk-24\bin\javaw.exe" on "Jul 24, 2026, 7:56:29 PM" with an elapsed time of "0:01:44" and PID "10688".

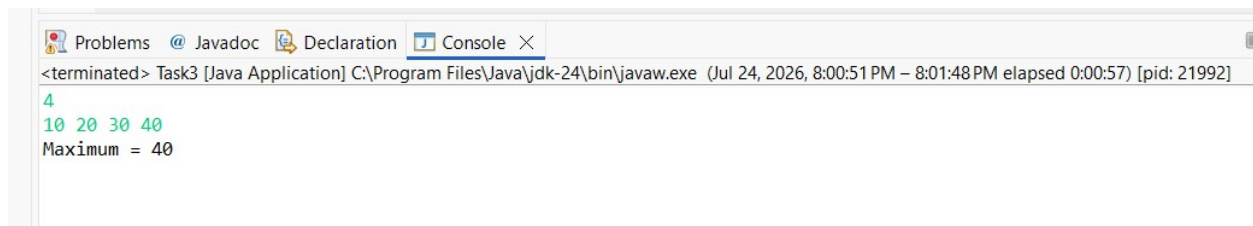
```
<terminated> Task2 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 7:56:29 PM – 7:58:13 PM elapsed 0:01:44) [pid: 10688]
3
10
12
30
37
Not Found
```

Task:3 Find the maximum element in an array of n integers

Program:

```
import java.util.*; import
java.util.stream.*; public
class Task3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int max = IntStream.range(0,n)
            .map(i->sc.nextInt())
            .max()
            .orElse(0);
        System.out.println("Maximum = "+max);
    }
}
```

Output:

A screenshot of an IDE's console window. The window has tabs for 'Problems', 'Javadoc', 'Declaration', and 'Console'. The 'Console' tab is active, showing the output of a Java application. The output consists of three lines: a green '4' on the first line, a green '10 20 30 40' on the second line, and 'Maximum = 40' on the third line. The window title bar indicates the application is 'Task3 [Java Application]' running on 'C:\Program Files\Java\jdk-24\bin\javaw.exe' with a PID of 21992.

```
<terminated> Task3 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 8:00:51 PM – 8:01:48 PM elapsed 0:00:57) [pid: 21992]
4
10 20 30 40
Maximum = 40
```

Task:4 Given an array of integers and a positive integer K, write a program to find :The Kth smallest element

Program:

```
import java.util.*;

import java.util.stream.*;

public class Task4 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n=sc.nextInt();

        List<Integer> list=IntStream.range(0,n)

        .map(i->sc.nextInt())

        .boxed()

        .sorted()

        .collect(Collectors.toList());

        int k=sc.nextInt();

        if(k<=list.size())

            System.out.println(list.get(k-1));

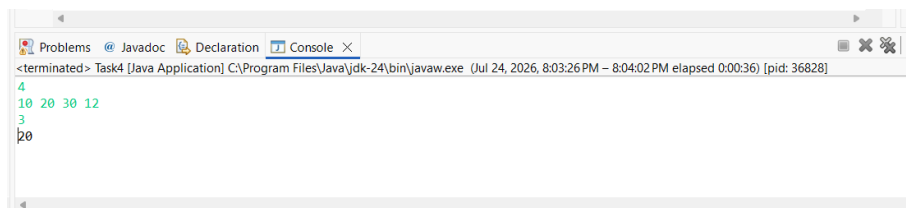
        else

            System.out.println("Invalid K");

    }

}
```

Output:



The screenshot shows a Java IDE console window with the following output:

```
<terminated> Task4 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 8:03:26 PM - 8:04:02 PM elapsed 0:00:36) [pid: 36828]
4
10 20 30 12
3
20
```

Task:5 Print all possible pairs of elements from an array of size n.

Program:

```
import java.util.*;
import java.util.stream.*;

public class Task5 {

    public static void main(String[] args)

    { Scanner sc=new

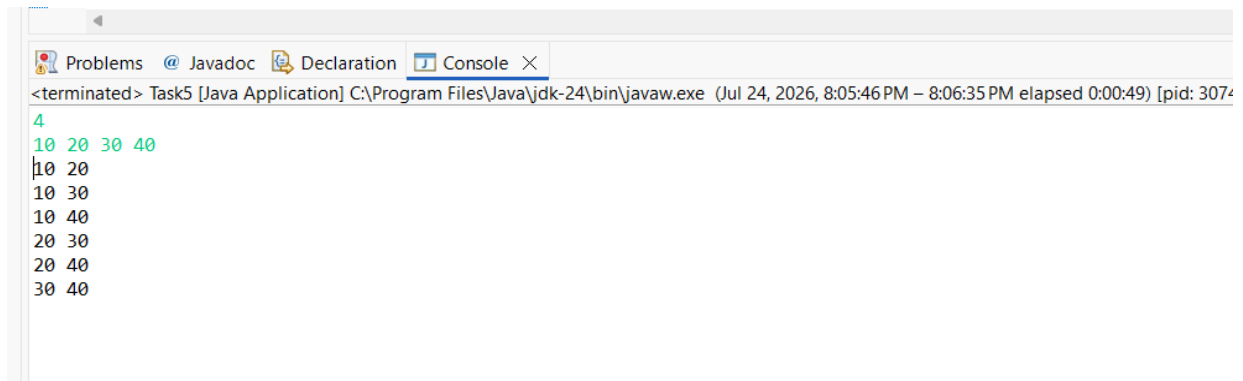
    Scanner(System.in); int

    n=sc.nextInt();

    List<Integer> list=IntStream.range(0,n)
        .map(i->sc.nextInt())
        .boxed()
        .collect(Collectors.toList());

    IntStream.range(0,n)
        .forEach(i->
            IntStream.range(i+1,n)
                .forEach(j->
                    System.out.println( list.get(
                        i)+" "+list.get(j)
                    )
                )
            )
        );
```

Output:



```
<terminated> Task5 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 8:05:46 PM – 8:06:35 PM elapsed 0:00:49) [pid: 3074]
4
10 20 30 40
10 20
10 30
10 40
20 30
20 40
30 40
```

Task-6: digitSum opt: sum of even or odd digits

Program:

```
import java.util.*;

import java.util.stream.*;

public class Task6 {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        String num=sc.next();

        int even=num.chars()

        .map(c->c-'0')

        .filter(d->d%2==0)

        .sum();

        int odd=num.chars()

        .map(c->c-'0')

        .filter(d->d%2!=0)

        .sum();

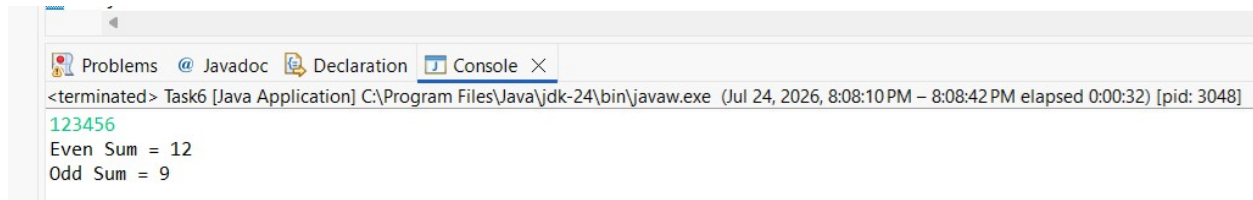
        System.out.println("Even Sum = "+even);

        System.out.println("Odd Sum = "+odd);

    }

}
```

Output:

A screenshot of a Java IDE's console window. The window has tabs for 'Problems', 'Javadoc', 'Declaration', and 'Console'. The 'Console' tab is active, showing the output of the program. The output starts with a green prompt '123456', followed by 'Even Sum = 12' and 'Odd Sum = 9'. The window title bar indicates the file path 'C:\Program Files\Java\jdk-24\bin\javaw.exe' and the execution time 'Jul 24, 2026, 8:08:10 PM - 8:08:42 PM elapsed 0:00:32 [pid: 3048]'.

```
<terminated> Task6 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 8:08:10 PM - 8:08:42 PM elapsed 0:00:32) [pid: 3048]
123456
Even Sum = 12
Odd Sum = 9
```


Task-7 Nth Fibonacci

Program:

```
import java.util.stream.*;

import java.util.*;

public class Task7 {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        int n=sc.nextInt();

        Stream.iterate(new int[]{0,1},

t->new int[]{t[1],t[0]+t[1]})

        .limit(n)

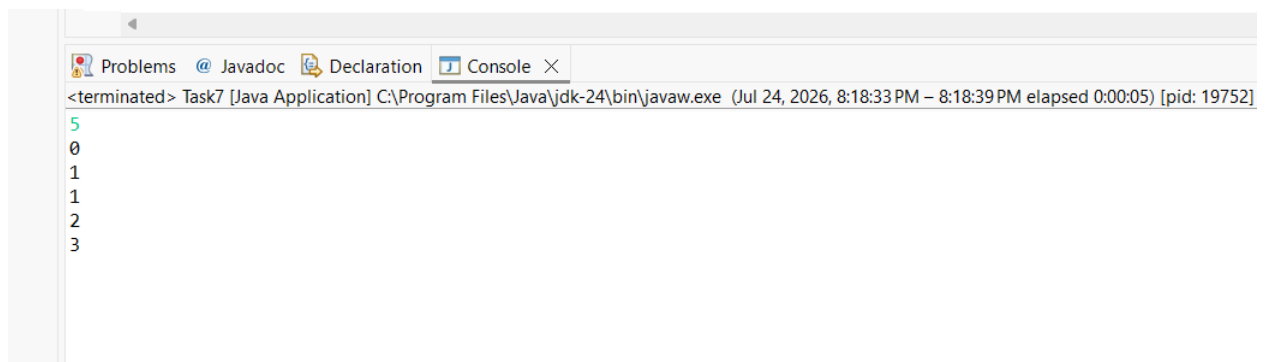
        .map(t->t[0])

        .forEach(System.out::println);

    }

}
```

Output:

A screenshot of a Java IDE's console window. The window has a title bar with tabs for 'Problems', 'Javadoc', 'Declaration', and 'Console'. The 'Console' tab is active, showing the output of a Java application. The output text is: '<terminated> Task7 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 8:18:33 PM – 8:18:39 PM elapsed 0:00:05) [pid: 19752]'. Below this, the numbers 5, 0, 1, 1, 2, 3 are printed on separate lines, representing the first six Fibonacci numbers. The console window has a light gray background and a vertical scrollbar on the right.

```
<terminated> Task7 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 8:18:33 PM – 8:18:39 PM elapsed 0:00:05) [pid: 19752]
5
0
1
1
2
3
```

Task-8: Is Palindrome Number?

Program:

```
import java.util.*;

public class Task8 {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        String num=sc.next();

        String rev=new StringBuilder(num)

            .reverse()

            .toString();

        System.out.println(num.equals(rev)?

            "Palindrome":

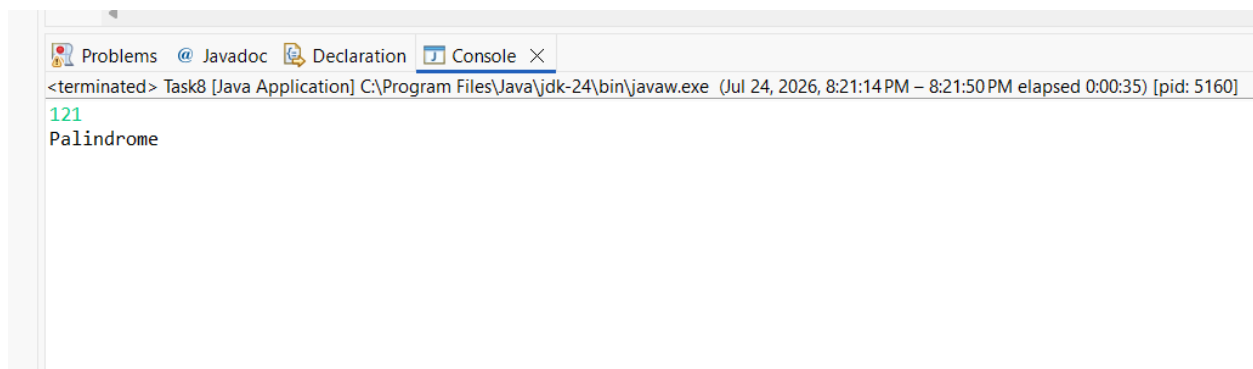
            "Not Palindrome"

        );

    }

}
```

Output:

A screenshot of an IDE's console window. The window has a title bar with icons for Problems, Javadoc, Declaration, and Console. The console text shows the program execution details: "<terminated> Task8 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 8:21:14 PM – 8:21:50 PM elapsed 0:00:35) [pid: 5160]". Below this, the line number "121" is shown in green, followed by the output "Palindrome".

```
<terminated> Task8 [Java Application] C:\Program Files\Java\jdk-24\bin\javaw.exe (Jul 24, 2026, 8:21:14 PM – 8:21:50 PM elapsed 0:00:35) [pid: 5160]
121
Palindrome
```

Task-9 Sum of last digit of two given numbers

Program:

```
import java.util.*;
import java.util.stream.*;

public class Task9 {

    public static void main(String[] args)

    { Scanner sc=new

    Scanner(System.in);

    int sum=Stream.of(sc.nextInt(),sc.nextInt())
        .mapToInt(n->n%10)
        .sum();

    System.out.println(sum);
    }
}
```

Output:



```
PS C:\Users\kthan\OneDrive\Desktop\Java programmings> & 'C:\Program Files\Java\jdk-24\bin\'
'-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\kthan\AppData\Roaming\Code\User\w
orpace\22d85c4fb85d0bbb796596baf7832d15\redhat.java\jdt_ws\Java programmings_a5db5e7c\bin' '
123
234
7
PS C:\Users\kthan\OneDrive\Desktop\Java programmings> 
```

END